

Targeted Research Synthesis: Balancing Differential Privacy Budgets with Diagnostic Accuracy in Federated Learning for Medical Imaging

I. Problem Definition

Federated Learning (FL) technology allows hospitals and other healthcare organizations to train their AI models together while keeping patients' private data intact. Despite all benefits, FL is still susceptible to information leakage through parameters exchange. That is why many scientists prefer using Differential Privacy (DP) in their work as the latter allows protecting sensitive information due to adding noise to model updates. Unfortunately, more robust privacy means lower accuracy of the models trained via FL.

This problem is especially important in the case of healthcare as AI models are used for diagnosing patients, analyzing medical images, making decisions about treatment, etc. Too much privacy noise can decrease the ability to diagnose diseases properly, thus leading to wrong positive or negative diagnosis that can influence patients' health. Too little privacy noise means increased risks of leaking private data and violating such healthcare legislation as HIPPA and GDPR. Thus, finding a balance between privacy and accuracy is one of the biggest issues in FL.

II. State-of-the-Art (SOTA) Analysis (2023–2026)

Recent healthcare AI studies focus on privacy, security, explain ability, and trustworthy deployment. Representative approaches are compared below.

Study	Year	Approach	Privacy	Accuracy	Latency/ Overhead	Key Limitation
Nguyen et al.	2023	Survey of privacy-preserving FL for smart healthcare	High	High	Medium	Limited practical deployment guidelines.
Li et al.	2024	Multi-stage Asynchronous FL with Adaptive Differential Privacy	Very High	High	Medium	Increased algorithmic complexity for asynchronous optimization.
Chen et al.	2024	Adaptive Differential Privacy Federated Learning (AWDP-FL)	Very High	High	Medium	Requires careful privacy budget tuning across clients.
Ahmed et al.	2024	Adaptive DP-enabled FL for COVID-19 diagnosis	High	High	Low	Evaluated mainly on COVID-19 chest X-ray datasets.
Pan et al.	2024	Adaptive FL framework for multi-hospital clinical risk prediction	High	Very High	Medium	Does not fully address communication cost in large healthcare networks.

III. Technical Gaps

- Although there have been some advances recently, there are still many technical gaps that need to be filled.
- Many algorithms in Differential Privacy use a constant privacy budget, which is not flexible enough for various medical data sets and clinical applications.

- Most algorithms focus on benchmark data set evaluation rather than implementation in a real multi-hospital environment.
- The integration of Differential Privacy with Secure Aggregation and Homomorphic Encryption significantly increases communication cost and computation cost.
- Personalized Federated Learning performs well locally but has difficulty scaling up in large medical networks.
- Current algorithms do not account for heterogeneous (non-IID) data from diverse medical institutions.

IV. Proposed Direction

One future research direction that seems promising is the design of an Adaptive Privacy-Aware Federated Learning (APFL) framework. In the new framework, the Differential Privacy budget would be flexibly adjusted according to the sensitivity of the data, the convergence of the model, and the performance of the clients. By combining the Secure Aggregation algorithm with dynamic allocation of privacy budgets, it is possible to keep the diagnostic accuracy high while minimizing the overhead of communication. Additionally, personalizing the models and employing efficient aggregation algorithms would make the framework scalable in diverse hospital settings.

V. Bibliography

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